CHLOÉ DUBRAY

Engineering graduate from École Centrale de Lyon | MSc in Bioinformatics and Modelling

Research internship — quantitative bioinformatics and modelling

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Profile Engineering graduate of École Centrale de Lyon and currently a second-year Master's student in Bioinformatics and Modelling at Sorbonne University. I have a broad scientific and mathematical background, with a specialisation in computer science and data science, as well as professional experience in software engineering and data processing. I am now moving towards research in bioinformatics, with a particular interest in population genetics comparative and evolutionary genomics, and the modelling of biological systems.

Availability : 5–6-month internship starting in mid-February 2027 | **Career goal :** pursue research, with the possibility of a PhD

MSc in Bioinformatics and Modelling

Sorbonne University, Paris · 2026–2027

Bilingual, interdisciplinary Master's programme run by the Computer Science and Molecular and Cellular Biology departments from Sorbonne University. Entered directly into the second year. Coursework : genetics and genomics, evolutionary biology, population genetics, statistical analysis of biological data, machine learning, biological networks and systems.

Sept. 2026–Present

Population Genetics — Simulation of Evolutionary Dynamics

Sorbonne University · Academic project

Developing population simulations to study the effects of genetic drift, mutation and selection; explore how demographic and evolutionary parameters affect allele frequencies; and compare the results with theoretical models.

Python · Stochastic simulation · Drift / Selection / Mutation

Oct. 2026–Present

Biological Networks — Inference of Immune Regulatory Networks

Sorbonne University · Academic project

Comparing immune regulatory networks in inflammatory bowel disease. Using transcriptomic data to infer networks and compare signalling pathways involved in inflammation in healthy and inflamed tissues. Data preparation, exploratory analysis, and reconstruction and comparison of gene association networks using statistical and information-theoretic network inference methods.

R · Transcriptomic data · Graph theory · ARACNE / PC / MIIC

Oct. 2026–Present

MeetU — Predicting the Impact of BRCA1 Variants

Sorbonne University / University of Warsaw · European collaborative project

Collaborative project investigating the impact of BRCA1 variants, in partnership with the University of Warsaw and CeNT (Center of New Technologies). Developing and training a classification model and an automated data pipeline to address a biological research question.

Python · Machine learning

Professional Experience

Oct. 2024–Dec. 2025

Data Engineer / Software Engineer

Advanced Schema, Paris · Permanent position · 14 months

- Data modelling; design, development and maintenance of relational and document databases.
- Developing Python scripts to automate and optimise data-processing pipelines.
- Designing a distributed platform for large-scale data processing : developing Java / Spring Boot services and APIs for data ingestion and access, and integrating processing workflows and business logic in a containerised cloud environment (Docker, Kubernetes).
- Software development, integration and deployment, including automated testing, code reviews, version control and process documentation, with a focus on code quality and maintainability.

Python · Java · Spring Boot / Reactor · SQL · Snowflake · SQL Server · MongoDB · Kafka · Git · Docker / Kubernetes

Apr. 2024–Sept. 2024

Software Developer

Devoteam, Vilnius, Lithuania · Final-year internship · 6 months

Developing Python APIs to make data from relational databases accessible, transforming and structuring it to meet business requirements. Exploring time-series forecasting methods using historical data to predict resource consumption.

Python · PySpark · FastAPI · SQLAlchemy · SQL · Azure Databricks · PostgreSQL · Git

May 2022–June 2023

Data Engineer / Cloud Architect

Cloud Temple, Paris · First-year Master's and gap-year internships · 14 months

Analysing technical requirements and designing information system architectures for tenders. Writing detailed technical documentation on architectures and their components : infrastructure, networks, computing, storage and cybersecurity.

Information systems · Technical documentation

Education

École Centrale de Lyon · 2020–2024

General engineering degree (Master's level), with a specialisation in computer science and data science. Strong academic foundation in applied mathematics, probability, statistics, algorithms and optimisation. Experience in modelling, literature reviews, report writing and the presentation of scientific work.

Jan. 2024–Apr. 2024

Optimisation under constraints — Public Transport Networks

École Centrale de Lyon · Final-year project

Formulating an optimisation problem to represent transport networks under graphical constraints and produce readable maps from a geographic network. Modelling the constraints, implementing and evaluating solutions.

Python · Geometric constraint optimisation · Graph theory

Jan. 2022–March 2022

ENS Data Challenge — Prediction of Molecular Properties

École Centrale de Lyon · Academic project

Predicting the properties of chemical compounds from their structure. Data preparation, supervised learning and model evaluation using random forests and neural networks.

Python · scikit-learn · Pytorch · Supervised learning

Sept. 2020–June 2021

Augmented Reality and Photon Cages

Lyon Nanotechnology Laboratory · Academic project

Working in a team to design an augmented reality application and science communication materials presenting research on photon cages to a scientific audience. Developed for the Lyon Nanotechnology Laboratory.

Laboratory collaboration · Science communication · Augmented reality

Preparatory Classes in Mathematics and Physics · Lycée Stanislas, Paris · 2018–2020

Intensive training in mathematics, physics and theoretical computer science in preparation for the competitive entrance examinations to French engineering schools.

Skills

Programming	Python · R · SQL · Java · Git
Modelling	Probability · Statistics · Stochastic simulation · Optimisation · Machine learning
Bioinformatics	Genomic and transcriptomic data · Population genetics · Biological networks · Graph theory
Languages	French (native) · Finnish (native) · English (fluent, TOEFL 653/677, 2022) · German (school level)